

D3: Direction-Robust Sensor Count and Layout

9 October 2026 (rerun)

Does a layout survive an unknown heading or a directional sea?

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SHARC Code: `runDirectionRobustLayoutStudy.m`, `d3Tools.m`, `directionalSpectralMatrix.m`, commit 62264f1

Results: `Results/D3/D3_20261009_135210`

Rerun after the amplitude amendment (8 Oct). The EMM FRFs used by the 5 Oct run were low by $\alpha(\omega) = H\omega^2/g$; `computeSensorFRF` now multiplies by α and the twin cache was migrated exactly (`cfg.meta.changes`, 2026-10-08). The D1 gate reference followed the D1 rerun (s18 to s6, logged 2026-10-08); settings otherwise unchanged; tester 14/14. This summary supersedes D3.20261005_155648. Three conclusions changed:

- Precision no longer limits any design: worst-case $\sigma_{\ln \beta} \leq 0.35\%$ even with two sensors.
- The registered count rule now recommends four sensors (was three), because the A-optimal layouts at $n_s = 2$ and 3 are Marginal on Welch bias.
- Placement now *can* meet the bias gate for a discrete secondary direction at $n_s = 4$. The spread sea is still not rescued.

Summary

Question. D1 found that two sensors suffice, but it assumed one *known* wave heading, and every optimum lay on the incidence axis. D3 removes that assumption in two separate ways.

- **D3a:** one heading, but unknown. The model is still correct, and the heading ψ is a nuisance parameter (Schur complement).
- **D3b:** a genuinely directional sea. The frozen single-direction model is wrong (V2), and each layout is judged on its own directional bias and covariance.

The search is exhaustive over all 528, 5456 and 40 920 layouts of 2, 3 and 4 sensors, 24 headings, with the best reference per condition.

Outcome.

- **D3a: an unknown heading costs geometry, not precision.** The worst-case $\sigma_{\ln \beta}$ of the robust optimum is 0.35, 0.097 and 0.069% for 2, 3 and 4 sensors, all far inside 1%. The class is set by the Welch bias. Under the registered rule (smallest n_s whose worst-case A-optimal layout is Identifiable at every heading) the recommendation is $n_s = 4$, at all three noise levels. Both added sensors pass the 20% rule (gains 72% and 29%). However, 28 two-sensor and 1760 three-sensor layouts are Identifiable at every heading: the rule's A-ranking does not pick them.
- **D3b: placement rescues discrete directions, not spread.** At $n_s = 4$, 460 to 6368 layouts pass the $d_{\text{tot}} \leq 0.5$ gate in each two-component sea (least-bad 0.20 to 0.37). At $n_s = 3$ the best is 0.58 to 0.87, and at $n_s = 2$ it is 21 to 39. The $\cos^{2s}$ spread sea stays at $d_{\text{tot}} = 25$ to 40 at every n_s , so no layout is feasible across all seas.

1 What the study does

Why D3 exists. D1 forced the floe from one known heading, so the transmissibility varied most along the incidence axis and every optimum lay on it. In the field the heading is estimated, not known (D3a), and the sea has spread or several components, which V2 showed breaks the single-direction model (D3b). The two problems are kept apart: in D3a the model is right and only information is lost; in D3b the model is wrong, and the question is whether a layout can make the error small enough.

The design is otherwise as D1: the 33-point candidate grid, p_0 , 0.10 m spacing, LSM6DSV16X headline with ICM-42688-P and G3 repeated, and the reference optimised per condition (a processing choice).

2 Method

Pre-registered in programme rev. 4 and `cfg.D3`; clarifications and one amendment logged before the first run; unchanged for the rerun apart from the gate reference.

Headings by symmetry. A heading ψ samples the nominal field at points rotated by $-\psi$. The grid repeats every 45° , so heading $45k + \delta$ of a layout equals heading δ of that layout rotated by $-45k$. One EMM field at the D1 grid and its copies at $\delta = 7.5, 22.5$ and 37.5° (plus $\pm 0.5^\circ$ copies for $\partial T / \partial \psi$) gives all 24 headings by index permutation. The tester confirms this against a copy sampled directly at 52.5° (2×10^{-12}).

Heading grid (amendment). The registered headings $0, 15, \dots, 345^\circ$ were replaced by the midpoint grid $7.5 + 15k^\circ$ before the first run. On a grid axis, a layout whose sensors all lie on that diameter (Level 2C, the D1 optimum) has $\partial T / \partial \psi = 0$ by mirror symmetry. Then $F_{\psi\psi} = F_{\theta\psi} = 0$ and the Schur complement is 0/0 at a measure-zero heading. The midpoint grid is the same uniform quadrature without that problem. Heading 0 is kept, with the heading known, as the D1 gate: it reproduces D1's two-sensor optimum (s6 + s18, reference s6; passed).

D3a: unknown heading as a nuisance. With $\vartheta = [\ln \beta, \ln R, \psi]$ and F partitioned into the $\theta = [\ln \beta, \ln R]$ and ψ blocks,

$$F_{\text{eff}} = F_{\theta\theta} - F_{\theta\psi} F_{\psi\psi}^{-1} F_{\psi\theta}, \quad A(\psi) = \text{tr } F_{\text{eff}}^{-1}.$$

Each layout is scored *per heading*, then by $\bar{A}$ (mean) and A_{worst} (max) over the 24 headings. A heading-averaged F is

never inverted, since that would describe an experiment repeated at every heading. The penalty is

$$\sigma_{\ln \beta}(\psi \text{ estimated}) / \sigma_{\ln \beta}(\psi \text{ known}).$$

D3b: directional seas. A second component at $\phi = 30, 60$ and 90° with $f_2 = 10$ and 25%, and one $\cos^{2s}$ spread ($s = 10, 24$ directions), each rotated with the primary heading over all 24 headings. The primary heading is known (isolating the model error from D3a), at the headline noise. For every layout, sea and heading:

- Σ from the rank-two (or spread) spectral matrix, and F and the projection weights under that Σ ;
- $b_{\text{dir}} = f_W^{\text{dir}} - f_W^{\text{single}}$;
- d_{tot} from $\delta\theta_W + \delta\theta_{\text{dir}}$.

Gate: a layout is feasible only if $d_{\text{tot}} \leq 0.5$ in every sea and heading, using for each condition the reference that passes with the smallest A . Choosing the reference with knowledge of the bias is optimistic, so a negative result is robust and a positive one is an upper bound on what placement achieves. The gate is never relaxed; the least-bad layout (smallest worst-case d_{tot}) is a diagnostic.

Count rule (as D1, on the worst-case optimum). A sensor is worth adding if it lowers the worst-case $\sigma_{\ln \beta}$ by $\geq 20\%$. The recommended count is the smallest n_s whose worst-case optimum (minimum A_{worst}) is Identifiable at every heading, and with D3b also bias-feasible.

3 Results: D3a, unknown heading

n_s	Design	Layout	worst-case $\sigma_{\ln \beta}$		mean	worst	efficiency
			ψ estimated	ψ known	$\sigma_{\ln \beta}$	class	mean / worst
2	D3a robust	s1 + s26	0.349%	0.320%	0.236%	Marginal	0.73 / 1.00
2	D3a mean-A	s3 + s31	0.360%	0.315%	0.198%	Marginal	1.00 / 0.93
2	D1 optimum	s6 + s18	0.627%	0.627%	0.231%	Marginal	0.62 / 0.31
3	D3a robust	s10 + s12 + s23	0.097%	0.096%	0.079%	Marginal	0.91 / 1.00
3	D3a mean-A	s6 + s11 + s33	0.104%	0.104%	0.076%	Marginal	1.00 / 0.86
3	D1 optimum	s6 + s18 + s28	0.120%	0.118%	0.083%	Marginal	0.81 / 0.65
4	D3a robust	s2 + s6 + s20 + s24	0.069%	0.069%	0.063%	Identifiable	0.96 / 1.00
4	D3a mean-A	s10 + s12 + s15 + s33	0.074%	0.074%	0.062%	Marginal	1.00 / 0.85
4	D1 optimum	s6 + s14 + s26 + s28	0.116%	0.109%	0.074%	Marginal	0.67 / 0.35
4	Level 2C	s2 + s6 + s14 + s26	0.164%	0.163%	0.101%	Identifiable	0.36 / 0.18

LSM6DSV16X. Worst/mean over the 24 headings, best reference per heading. Efficiency: the robust optimum's $\bar{A}$ (or A_{worst}) over the design's. Every $\sigma_{\ln \beta}$ is far below 1%, so "Marginal" here means a Welch bias $0.5 < d_W \leq 1$ at some heading. Layouts Identifiable at every heading: 28 of 528, 1760 of 5456, 14 988 of 40 920 (pre-amendment 0, 740, 21 760).

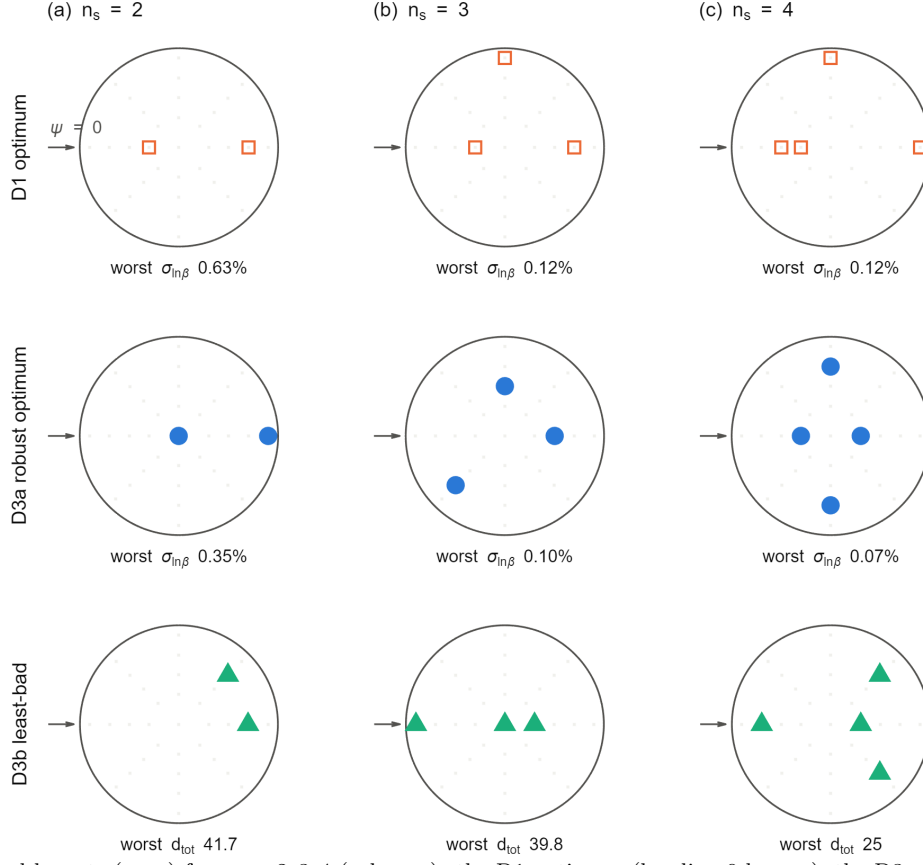


Figure 1. Selected layouts (rows) for $n_s = 2, 3, 4$ (columns): the D1 optimum (heading 0 known), the D3a robust (worst-case) optimum (heading unknown) and the D3b least-bad layout (smallest worst-case d_{tot} over all seas and headings, set by the spread sea). Grey: the candidate grid; arrow: heading $\psi = 0$.

The robust layouts.

- The robust two-sensor design pairs the centre (s1, whose response does not depend on heading) with one rim point.
- The robust four-sensor design is a symmetric cross.

The D1 axis layouts lose much less than before. D1's three-sensor optimum now includes an off-axis point (s6 + s18 + s28), and keeps 65% of the robust worst-case efficiency (pre-amendment: 3%). Level 2C, a pure axis layout, keeps 18% but is still Identifiable at every heading (0.164%).

Noise	worst-case optimum, worst $\sigma_{\ln\beta}$			gain		recommended n_s (rule)
	$n_s = 2$	3	4	2 → 3	3 → 4	
LSM6DSV16X	0.349%	0.097%	0.069%	72%	29%	4
ICM-42688-P	0.458%	0.117%	0.086%	74%	26%	4
G3 assumption	0.444%	0.115%	0.083%	74%	27%	4
pre-amendment, LSM	2.80%	0.60%	0.43%	79%	28%	3

Worst-case optimum classes at all three noise levels: Marginal, Marginal, Identifiable for $n_s = 2, 3, 4$.

Why the rule now asks for four sensors. Before the amendment the worst-case optimum at $n_s = 3$ was Identifiable on precision. Now every σ is tiny, and the class is set by the Welch bias at the worst heading. The rule ranks layouts by A (precision) and only then reads the class, so it picks the most precise layout even when a slightly less precise one is bias-free. At $n_s = 2$ and 3 those bias-free layouts exist (28 and 1760 Identifiable at every heading); the A-optimal ones are not among them. The rule is applied as registered. A longer Welch segment, which D2 and R2 found lowers d_W about threefold, would probably make the A-optimal $n_s = 2$ and 3 layouts Identifiable too, but that was not tested here.

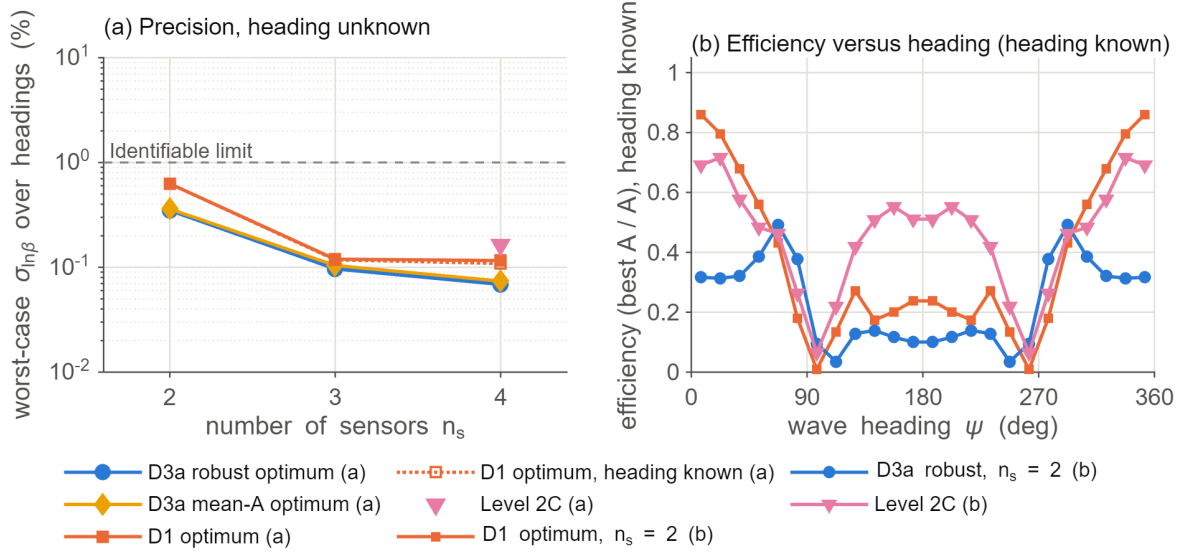


Figure 2. (a) Worst-case $\sigma_{\ln\beta}$ over the headings (heading estimated) against n_s for the D3a optima, the D1 optimum (also with the heading known) and Level 2C, with the 1% Identifiable limit. (b) Efficiency against heading with the heading known: best A at that heading over the design's A .

Geometry, not the heading estimate, is the cost. For the robust layouts the nuisance penalty is small:

- $n_s = 2$: 0.349% against 0.320% (+9%);
- $n_s = 3$: 0.097% against 0.096%;
- $n_s = 4$: none.

For individual two-sensor designs the penalty reaches 2.6 to 2.7 at some headings (Figure 3a). The D1 two-sensor axis layout is the most exposed: its efficiency falls to about zero near 90° (Figure 2b).

4 Results: D3b, directional seas

Sea	least-bad worst-case d_{tot} over 24 headings (precision-optimal reference)			layouts passing at $n_s = 4$
	$n_s = 2$	$n_s = 3$	$n_s = 4$	
$\phi = 30^\circ, f_2 = 10\%$	22.5 (24.4)	0.743 (1.24)	0.366 (0.524)	1034
$\phi = 30^\circ, f_2 = 25\%$	39.1 (43.3)	0.873 (1.47)	0.352 (0.594)	460
$\phi = 60^\circ, f_2 = 10\%$	22.6 (24.9)	0.624 (0.987)	0.200 (0.440)	6368
$\phi = 60^\circ, f_2 = 25\%$	38.9 (43.3)	0.712 (1.20)	0.257 (0.512)	3552
$\phi = 90^\circ, f_2 = 10\%$	21.0 (24.7)	0.582 (0.811)	0.299 (0.434)	5980
$\phi = 90^\circ, f_2 = 25\%$	36.7 (43.1)	0.677 (0.910)	0.370 (0.539)	1288
$\cos^2 s$ spread, $s = 10$	31.5 (33.0)	39.8 (49.1)	25.0 (31.8)	0
All seas (one layout)	41.7 (s18 + s19)	39.8 (s1 + s2 + s30)	25.0 (s2 + s19 + s22 + s25)	0

Per sea: the minimum over layouts (a different layout may achieve each). Gate $d_{\text{tot}} \leq 0.5$; no layout passes at $n_s = 2$ or 3 in any sea. In brackets: with the precision-optimal reference instead of the bias-aware one. Pre-amendment least-bad at $n_s = 4$: 0.97 to 1.34 (two-component), 24.9 (spread), 0 passing.

Placement now rescues discrete secondary directions. For the two-component seas the least-bad d_{tot} falls from 21 to 39 at $n_s = 2$, to 0.58 to 0.87 at $n_s = 3$, and to 0.20 to 0.37 at $n_s = 4$. There, hundreds to thousands of layouts pass each sea. With the precision-optimal reference, three of the six seas still pass at $n_s = 4$ (0.43 to 0.44), so the result does not rest entirely on the optimistic reference choice. Before the amendment no sea was passed at any n_s . The run does not diagnose why the best layouts now cancel the directional bias better. The change in bin weighting (as in R2) is the likely cause.

The run stores the per-sea pass counts but not their intersection, so it is not known whether one four-sensor layout passes all six two-component seas at once.

The spread sea is not rescued: 31.5, 39.8 and 25.0 for $n_s = 2, 3, 4$, essentially unchanged from before. With every heading component present, no sensor geometry separates the directions. Because the spread sea dominates every layout's worst case, the least-bad layouts in Figure 1 are spread-sea layouts.

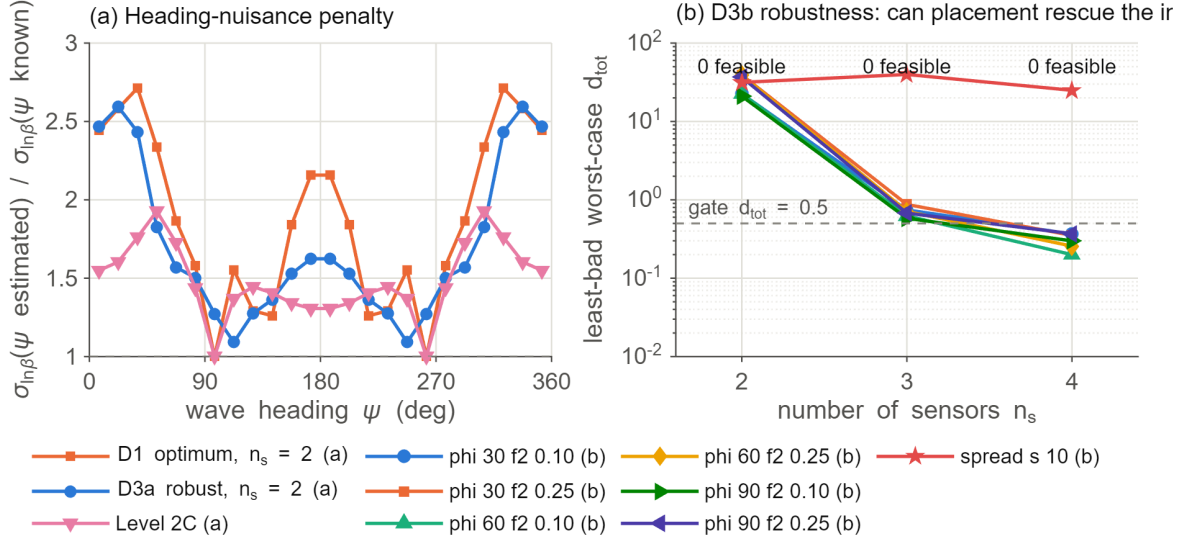


Figure 3. (a) Heading-nuisance penalty $\sigma_{\ln\beta}(\psi \text{ estimated})/\sigma_{\ln\beta}(\psi \text{ known})$ against heading for two-sensor designs and Level 2C. (b) D3b: least-bad worst-case d_{tot} per sea against n_s , with the $d_{\text{tot}} = 0.5$ gate. “0 feasible” refers to all seas at once; every two-component sea is passed at $n_s = 4$.

5 Interpretation

1. **An unknown heading no longer threatens precision.** Even the best two-sensor layout has worst-case $\sigma_{\ln\beta} = 0.35\%$. Before the amendment, two sensors could not be Identifiable at every heading. D1’s two-sensor result is now conditional on the bias, not on precision.
2. **The sensor count is now decided by the Welch bias.** The registered rule gives $n_s = 4$, and both extra sensors pass the 20% rule. But 28 two-sensor and 1760 three-sensor layouts are Identifiable at every heading. A bias-aware ranking, or a longer Welch segment, would lower the count. The thesis should report the rule’s answer and this caveat together.
3. **Estimating the heading is cheap; the layout’s shape still matters.** The Schur penalty is at most 9% for the robust optima. Axis layouts keep 18 to 65% of the robust worst-case efficiency (3 to 9% before the amendment).
4. **Placement can now make the inverse consistent under one discrete secondary direction.** This needs four sensors (with d_{tot} down to 0.2), and three come close (0.58 to 0.87).
5. **Placement cannot fix a spread sea.** $d_{\text{tot}} \approx 25$ to 40 at every n_s . For a realistically spread sea the frozen single-direction inverse remains structurally wrong.

Conclusion (thesis wording). D1 showed that two sensors suffice when the heading is known. With a single but unknown heading, precision remains excellent at every sensor count (worst-case $\sigma_{\ln\beta} = 0.35, 0.10$ and 0.07% for two, three and four sensors). The Identifiable class is decided instead by the Welch bias at the worst heading. Under the registered rule this gives four sensors, with each added sensor reducing the worst-case uncertainty by more than 20%, although many two- and three-sensor layouts are also Identifiable at every heading. The penalty for estimating the heading is at most 9% once the geometry is robust. Under directional seas, four well-placed sensors bring the frozen inverse within the $d_{\text{tot}} \leq 0.5$ criterion for every tested discrete secondary component, and three come close. No layout with up to four sensors is consistent under a spread sea. Directional spread must therefore be detected and excluded or represented in the inference model; it is not a sensor-placement problem.

Limitations.

- 33 discrete candidate points, $n_s \leq 4$, and one floe (p_0).
- D3a samples 24 headings. The heading derivative is a $\pm 0.5^\circ$ central difference, which changes A by at most 6×10^{-4} when halved.
- D3b runs at the headline noise only, with the primary heading known (D3a and D3b effects are not combined). There are seven registered seas, one of them spread ($s = 10$).
- The bias-aware reference choice is optimistic. The intersection of passing layouts across the six two-component seas was not stored.
- d_{tot} is a precision-scaled measure, and the classes use $L = 2048$.
- Apart from the forcing, truth and inverse share the EMM, including its truncation (V1).

Requirement for the experiment and the thesis

Unknown heading. Use angularly distributed sensors rather than an incidence-axis line. By the registered rule, four sensors (the cross $s_2 + s_6 + s_{20} + s_{24}$) are recommended. Two or three sensors give ample precision (worst case at most 0.35%) and can be Identifiable at every heading if chosen with the bias in mind. Do not rely on the D1 two-sensor axis design unless the heading is known.

Directional seas. Four well-placed sensors can keep the frozen inverse consistent under one discrete secondary direction. No layout copes with directional spread. Spread must be detected and excluded (V2: coherence, with a calibrated noise floor) or represented in the inference model.

Output (Results/D3/)	Contents
D3_20261009_135210.mat / _summary.txt	results and config; headline numbers (run 1874 s)
D3_d3a.csv, D3_d3b.csv, D3_counts.csv	D3a designs per noise and n_s ; D3b per design and sea; count rule
figures/D3_fig1 to D3_fig3 (.pdf/.png)	Figures 1 to 3 (plotDirectionRobustLayoutStudy)
Superseded	D3_20261005_155648 (pre-amendment)